

Modern machine learning systems based on the transformer architecture implement distributed computation through the exchange of information across distinct representation streams. While this information flow is central to model behavior, it is typically studied indirectly through geometric or representational analyses of trained models, rather than treated as an explicit object of control. This project introduces a complementary, communication-theoretic perspective in which internal information exchange is treated as a communication channel with tunable capacity that can be directly measured and regulated during learning.

We develop a controllable family of transformer models in which attention-mediated communication is constrained through variational information bottlenecks applied to residual stream updates (Fig.). This induces a continuous spectrum of models parameterized by communication capacity, ranging from independent local processing to fully coordinated global computation. Within this framework, internal information flow becomes a measurable and tunable quantity, enabling systematic study of how distributed computation emerges under constraint.

Using this spectrum as an experimental and theoretical scaffold, we will (i) establish how architectural design choices shape the semantics of constrained communication, (ii) identify stable communication primitives and circuit structures that persist across capacity regimes, and (iii) determine how data structure and learning objectives govern the emergence of coordinated information routing under capacity constraints.

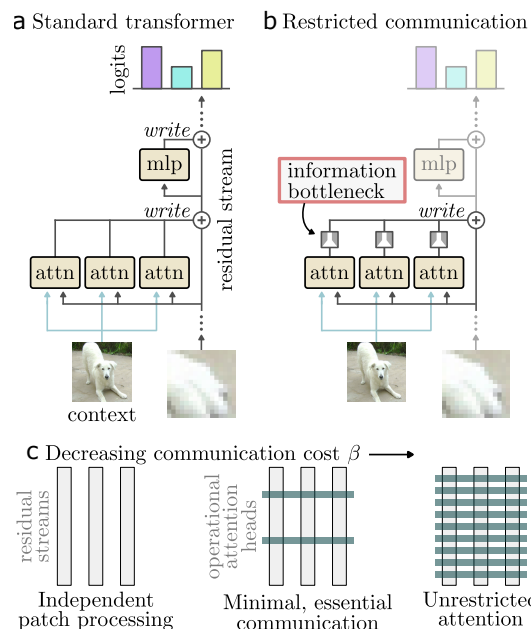
Together, this work provides a foundation for analyzing and designing information processing systems in which communication is explicitly controlled, yielding both new theoretical insights into distributed learning and practical tools for constructing controllable and interpretable models.

Updates since Feb. 2026 submission

- Extended framework to self-supervised learning (DINO), demonstrating that communication-constrained training applies beyond classification settings.
- Developed additional diagnostics for characterizing communication structure in high-capacity regimes (layerwise dynamics, cross-head statistics, activation patching).

Questions for Program Director

- The proposal was moved from RI to CIF during initial consideration. Are there specific aspects of the proposal that should be reframed to better align with CIF priorities?
- How important is it to emphasize connections to classical settings (e.g., multi-terminal information theory, rate-distortion) versus modern ML formulations?
- Would reviewers expect theoretical guarantees or formal limits, or is a well-characterized empirical framework sufficient at this stage?
- Is the use of transformers as a primary testbed appropriate for CIF, or should the proposal more explicitly abstract beyond this architecture?



Information bottlenecks applied to attention writes define a controllable communication channel with capacity governed by β , inducing a spectrum of distributed computation regimes from local processing to global coordination.